

FIN — Release 10.33

Research Programs P379–P393

Proof-reflected certificates, global radial injectivity,
noisy dynamics, and a first explicit operational realization

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Near-contact oscillatory certificate.

$$0.7073534379053974 \leq \mathcal{N}_{\text{osc}} \leq 0.7073534683998260, \quad \text{width} < 3.05 \times 10^{-8}.$$

Global strict radial theorem.

$$d \neq e, \quad d, e \in \mathbb{N}_0 \quad \implies \quad K_{\text{strict}}(d) \neq K_{\text{strict}}(e).$$

Thirty-five exact terminal predicates are checked by Lean 4.28. One positive-probability mathematical realization is constructed for the signed resource. No selector, dimensional standard, calibrated laboratory realization, role transfer, Standard Model, gravity, or Theory-of-Everything closure is claimed.

Repository. <https://github.com/hyconiek/Fractal-Nadsoliton-Theory>

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Publication statement

This monograph reports every executed Program P379–P393 and recommends Programs P394–P410. Exact proof, proof-kernel reflection, bounded numerical evidence, conditional operational construction, and absent external evidence are kept distinct. The legacy/strict split and all current strict-core guardrails remain binding.

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Confidence convention

Label	Meaning
[Proven]	Mathematical proof, exact rational computation, proof-kernel-checked terminal predicate, or exhaustive finite result
[Strong evidence]	Reproducible high-precision or bounded finite computation with explicit limitations
[Moderate evidence]	Model-dependent synthetic or local result
[Conditional]	Requires named preparations, encodings, standards, calibrations, or axioms
[Refuted]	The declared proposition fails in its stated class
[Blocked by external evidence]	Independent data, hardware, calibration, custody, standards, or unblinding is required

Chapter 1

Scientific scope and guardrails

1.1 The kernel split remains binding

The legacy intermediate kernel is

$$K_{\text{legacy,ont}}(d) = 4 \ln 2 \frac{\cos(\pi d/4 + \pi/6)}{1 + 0.01d},$$

whereas the strict working kernel is

$$K_{\text{strict,gate}}(d) = \frac{\cos((743/4000)d + 13/80)}{1 + d^{9/5}}.$$

P382 completes only the attenuation denominator after amplitude and phase have been separated. It is not an identity between the two full kernels.

1.2 Ontology

The repository order remains

$$\text{nadsoliton} \longrightarrow \text{light} \longrightarrow \text{matter} \longrightarrow \text{emergent observer}.$$

The nadsoliton is the primordial information in a solitonic state. No proof in this report assumes another informational substrate below it.

1.3 Preserved open boundaries

The following remain open:

- QW-2191 and a non-premise selector;
- a source-independent complete legacy-to-strict map;
- transfer of legacy electroweak or hierarchy roles;
- internal length, time, action, mass, or energy standards;
- calibrated laboratory preparation and measurement;
- a unit-bearing nonproxy total action;
- Standard Model or gravity generation;
- Theory-of-Everything closure.

Chapter 2

Executive result table

Program	Result	Status
P379	35 exact terminal predicates checked by Lean	[Proven] / end-to-end reflection open
P380	Oscillatory gap reduced to 3.05×10^{-8}	[Proven]
P381	Strict kernel globally injective on $\mathbb{N}_0$	[Proven]
P382	One damping naturality square; full completion fails	[Proven] / [Refuted]
P383	Exact $(2\pi\mathbb{Z})^{66}$ coordinate gauge	[Proven] / global aliases open
P384	No calibrated photonic pilot admitted	[Blocked]
P385	Noisy adaptive upper and parallel lower bounds	[Proven] / optimum open
P386	Clock-tube discrimination envelope	[Proven] / [Conditional]
P387	Jordan Sampling Realization functor	[Proven] / physical execution conditional
P388	Twenty forced cross-conversions refuted	[Proven]
P389	Dilation semigroup does not select ρ	[Proven]
P390	No independent QW hold-out admitted	[Blocked]
P391	No dimensional standards bundle admitted	[Blocked]
P392	No reservoir tomography bundle admitted	[Blocked]
P393	No electroweak blind-test bundle admitted	[Blocked]

Chapter 3

P379 — what proof-kernel reflection now means

3.1 Reflected predicates

The generated Lean source contains:

Predicate class	Count
Krawczyk lower/upper containment	24
Certified weight signs	7
Bernstein lower/upper bounds	2
Weak-duality order and contact gap	2
Total	35

Each rational comparison

$$\frac{a}{b} < \frac{c}{d}$$

is converted to an integer proposition

$$ad < cb$$

with positive denominators. Lean evaluates the exact integer proposition by kernel-checked decision.

3.2 What is no longer trusted

A silent floating comparison is not trusted. Decimal display values are not the proof objects. Hash identity alone is not the proof object. The terminal containment, sign, feasibility, order, and gap statements are represented by exact integers.

3.3 What is still trusted

The following generators remain Python implementations:

- Taylor/Lagrange cosine intervals;
- fifth-root brackets;
- interval matrix arithmetic;
- Krawczyk construction;
- Bernstein subdivision;
- conversion from archived mathematical inputs to terminal rational claims.

Therefore P379 is a genuine reduction of the trusted base, not a complete formalization of the numerical proof pipeline.

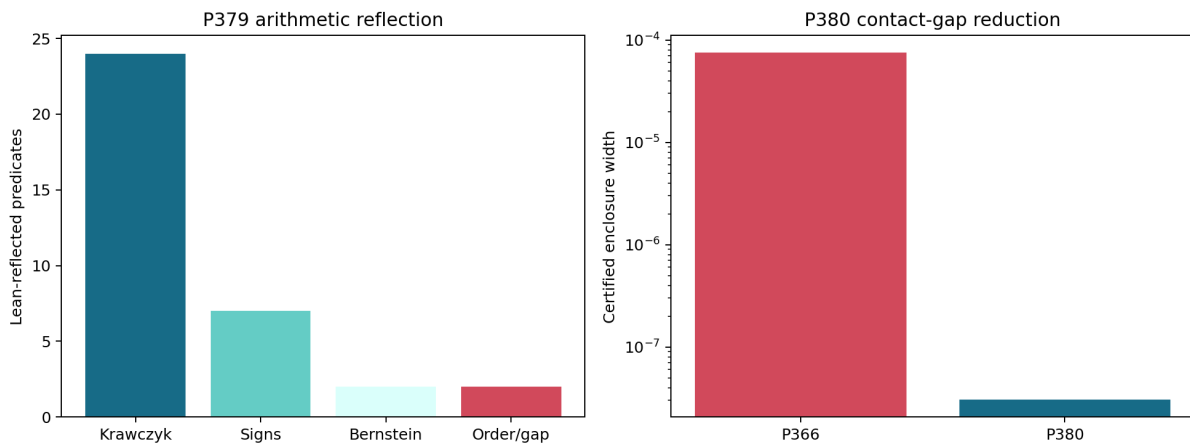


Figure 3.1: Lean-reflected terminal predicates and the reduction of the exact oscillatory primal-dual gap.

Chapter 4

P380 — near-contact oscillatory extremality

4.1 Why the old basis was inefficient

The raw degree-eleven monomial representation contains coefficients of very different magnitudes. Feasibility is a statement about a bounded function, but monomial coefficients are a poorly conditioned coordinate system for that statement.

P380 optimizes and freezes the polynomial in the Bernstein basis. Local Bernstein coefficients on dyadic cells directly bound the polynomial.

4.2 Exact normalization

Let $b_0, \dots, b_{11} \in \mathbb{Q}$ be the frozen Bernstein coefficients. On all 2^{14} dyadic cells, exact de Casteljau subdivision gives rational extrema L, U . Define

$$p_{\text{safe}}(x) = \frac{p_{\text{raw}}(x) - U}{U - L}.$$

Every local Bernstein coefficient then belongs to $[-1, 0]$. Therefore

$$-1 \leq p_{\text{safe}}(x) \leq 0 \quad (0 \leq x \leq 1).$$

4.3 Primal-dual result

The exact dual lower endpoint is

$$0.7073534379053974.$$

The existing exact seven-atom Krawczyk primal gives

$$0.7073534683998260.$$

Weak duality proves the global enclosure stated in the abstract.

4.4 Interpretation

The result establishes a nearly closed extremal problem in classical signed Hausdorff moment theory. It does not choose a physical ontology for signed mass. P387 supplies one positive-probability operational encoding, but it is not forced uniquely.

Chapter 5

P381 — the global strict-distance theorem

5.1 Statement

For

$$k(d) = \frac{\cos(x_d)}{1 + d^{9/5}}, \quad x_d = \frac{743d + 650}{4000},$$

the restriction

$$k : \mathbb{N}_0 \longrightarrow \mathbb{R}$$

is injective.

5.2 Proof

For every $d \in \mathbb{N}_0$, x_d is a positive rational and

$$a_d = \frac{1}{1 + d^{9/5}}$$

is a nonzero algebraic number.

Assume $d \neq e$ and $k(d) = k(e)$. Then

$$a_d(e^{ix_d} + e^{-ix_d}) - a_e(e^{ix_e} + e^{-ix_e}) = 0.$$

The four exponents

$$ix_d, \quad -ix_d, \quad ix_e, \quad -ix_e$$

are distinct algebraic numbers. The Lindemann–Weierstrass theorem says that their exponentials are linearly independent over the algebraic numbers. The displayed nontrivial relation is impossible.

Therefore $k(d) \neq k(e)$.

5.3 Categorical consequence

P368 identified the maximal radial naturality category as maps preserving distance modulo equality of kernel values. P381 removes the quotient ambiguity globally:

$$k(d_H(fx, fy)) = k(d_G(x, y)) \iff d_H(fx, fy) = d_G(x, y).$$

This conclusion holds on integer shortest-path metrics of arbitrary finite diameter.

5.4 Boundary

The theorem is formula-specific. It does not say that the same function is natural on continuous metric spaces, nor that the legacy kernel shares its category, nor that physical spacetime is a graph metric.

Chapter 6

P382 — one bridge atom, and no more

6.1 The exact square

Define

$$a_L(d) = \frac{1}{1 + 0.01d}, \quad a_S(d) = \frac{1}{1 + d^{9/5}}.$$

Then

$$C_{\text{damp}}(d)a_L(d) = a_S(d)$$

identically.

For an isometric embedding $f : G \rightarrow H$,

$$C_{\text{damp}}(d_H(fx, fy)) = C_{\text{damp}}(d_G(x, y)).$$

Thus pullback commutes with entrywise damping completion.

6.2 Why this is not a source theorem

The formula for C_{damp} already contains:

- the strict unit coefficient;
- the strict exponent $9/5$;
- the legacy coefficient 0.01 .

It packages a target comparison. It does not derive strict compression from legacy data.

6.3 Full-kernel falsification

After amplitude absorption and damping replacement, the legacy phase remains

$$\cos(\pi d/4 + \pi/6),$$

not

$$\cos((743/4000)d + 13/80).$$

The C_{12} relative profile residual is approximately 0.9944. Damping alone therefore fails as a full completion.

Chapter 7

P383–P384 — photonic gauge and physical boundary

7.1 Exact coordinate gauge

Each component phase appears only through $e^{\pm i\varphi_j}$. Hence

$$\varphi_j \sim \varphi_j + 2\pi n_j, \quad n_j \in \mathbb{Z}.$$

The natural canonical chart is

$$\mathbb{R}_+^{66} \times (-\pi, \pi]^{66}.$$

7.2 Finite audit

For all 66 components:

- a single 2π shift has maximum transfer defect below 8.0×10^{-16} ;
- 32 random combined phase-lattice shifts have maximum defect below 3.8×10^{-15} ;
- every tested single π shift has defect at least 0.285;
- every tested 0.01 component-loss shift has defect at least 4.99×10^{-3} .

These numbers identify coordinate gauges, not every possible distant compensating alias.

7.3 External boundary

P384 finds no admitted calibrated pilot. A physical pilot must supply complex transfer calibration, detector response, timestamps, preparation records, independent custody, a frozen protocol, and an unmodified hold-out report.

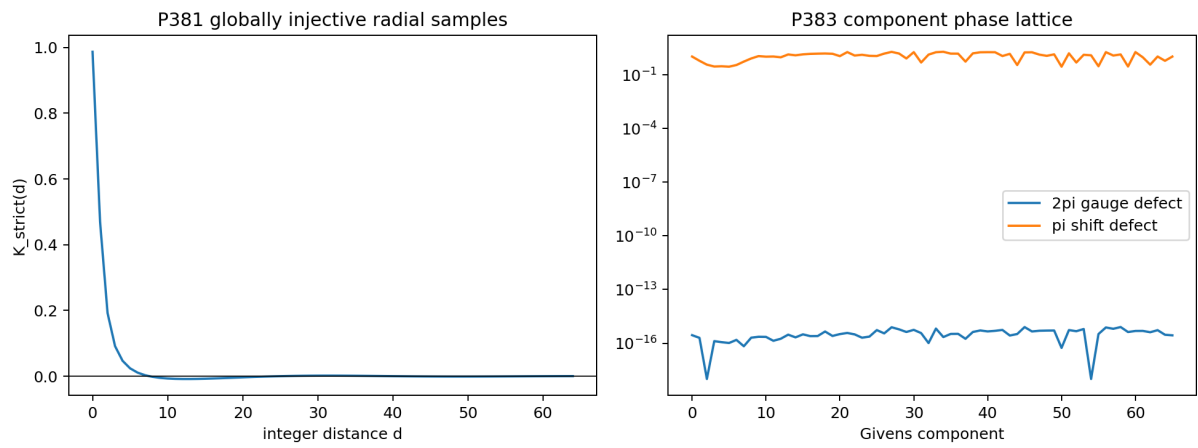


Figure 7.1: Global strict radial injectivity diagnostics and the photonic component-phase coordinate gauge.

Chapter 8

P385 — the noisy-comb sandwich

8.1 Declared reduced channel

Restrict to the two extremal eigenmodes of the relative generator. Let:

- $\eta \in [0, 1]$ be independent heralded survival per call;
- $q \in [0, 1]$ be eigenbasis coherence retention;
- $\theta = t\Delta/2$.

The two one-use states have off-diagonal phases $e^{\pm i\theta}$, attenuated by q . Orthogonal erasure flags contribute only on surviving branches.

8.2 One-use distance

The exact one-use half-distance is

$$\delta_1 = \eta q |\sin \theta|.$$

8.3 Adaptive upper bound

The hybrid/telescoping argument replaces oracle calls one at a time. Trace-distance contractivity under common adaptive controls gives

$$D_n^{\text{adaptive}} \leq \min(1, n\delta_1).$$

8.4 Feasible lower bounds

For independent probes, condition on k surviving calls:

$$L_{\text{prod}} = \sum_{k=1}^n \binom{n}{k} \eta^k (1-\eta)^{n-k} \frac{1}{2} \left\| \rho_+^{\otimes k} - \rho_-^{\otimes k} \right\|_1.$$

For the GHZ strategy, only the all-surviving coherent branch is used:

$$L_{\text{GHZ}} = \eta^n q^n |\sin(n\theta)|.$$

Therefore

$$\max(L_{\text{prod}}, L_{\text{GHZ}}) \leq D_n^{\text{adaptive}} \leq \min(1, n\eta q |\sin \theta|).$$

In the ideal case $\eta = q = 1$, the GHZ lower recovers the P371 first-branch optimum.

8.5 Open problem

The largest gap in the declared audit is approximately 0.5591. Closing it requires either:

- a channel-comb SDP with a dual certificate;
- an erasure-tolerant entangled code;
- a sharper adaptive contraction theorem;
- a proof that one of the displayed strategies is optimal in a restricted noise class.

Chapter 9

P386 — clocks deform predictions into tubes

If the true dimensionless time lies in

$$\tau \in [\hat{\tau} - \epsilon, \hat{\tau} + \epsilon]$$

and the interval remains on the first P371 branch, monotonicity gives

$$\sin\left(\frac{n\Delta(\hat{\tau} - \epsilon)}{2}\right) \leq D_n(\tau) \leq \sin\left(\frac{n\Delta(\hat{\tau} + \epsilon)}{2}\right).$$

Endpoints are clipped to the physical branch $[0, \pi/(n\Delta)]$.

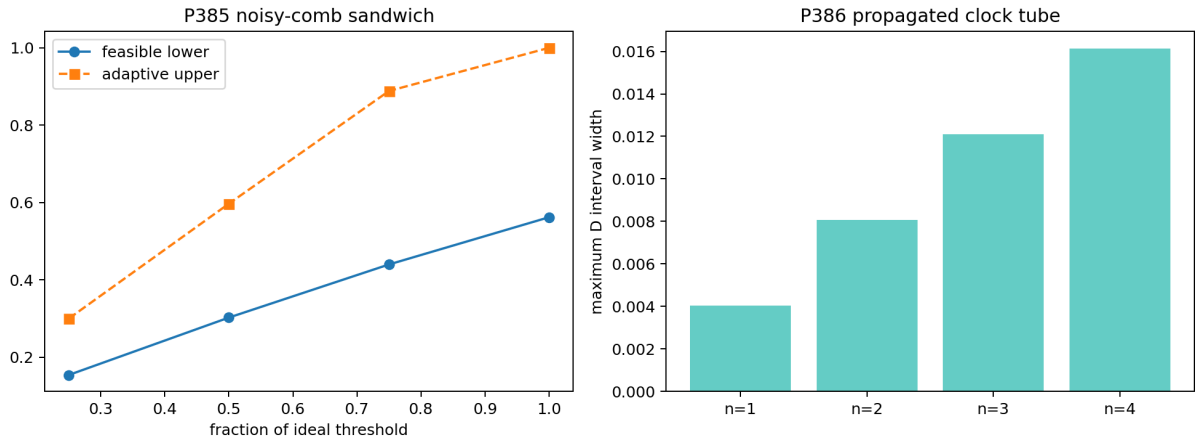


Figure 9.1: The declared noisy-comb lower/upper sandwich and propagated clock-calibration tubes.

The result separates two questions:

1. what FIN predicts after a clock map is supplied;
2. how the clock map is calibrated experimentally.

Only the first is solved here.

Chapter 10

P387 — Jordan Sampling Realization

10.1 Category of source objects

The source objects are finite signed measures

$$\mu = \sum_i w_i \delta_{x_i}.$$

Positive column-stochastic maps act on the signed weight vector.

10.2 Operational preparation

Define

$$V = \sum_i |w_i|, \quad q_i = \frac{|w_i|}{V}, \quad s_i = \text{sgn}(w_i).$$

The preparation distribution q is ordinary and positive.

10.3 Instrument and record

One trial produces

$$(i, x_i, s_i, \text{run id}, \text{timestamp}).$$

For mathematical reconstruction, the minimal record is (i, x_i, s_i) . Laboratory timestamps and run identifiers are required for custody and drift tests.

10.4 Estimator identity

For any function f ,

$$\hat{f} = V s_i f(x_i)$$

is unbiased:

$$\mathbb{E}_q[\widehat{f}] = \sum_i w_i f(x_i).$$

In particular, $f(x) = x^k$ reconstructs the signed moments.

10.5 Resource monotonicity

The Jordan negative mass is

$$\mathcal{N}(\mu) = \frac{\|\mu\|_{\text{TV}} - \mu(X)}{2}.$$

For a positive mass-preserving Markov map T ,

$$\|T\mu\|_{\text{TV}} \leq \|\mu\|_{\text{TV}}, \quad (T\mu)(X) = \mu(X).$$

Therefore

$$\mathcal{N}(T\mu) \leq \mathcal{N}(\mu).$$

10.6 What has and has not been constructed

Constructed mathematically:

- state/preparation;
- free dynamics;
- instrument;
- score;
- record schema;
- monotone;
- reconstruction theorem.

Not constructed physically:

- a device realizing the nodes and probabilities;
- a dimensional clock;
- detector calibration;
- environmental model;
- independent raw-event custody;
- a theorem that nature uses this realization.

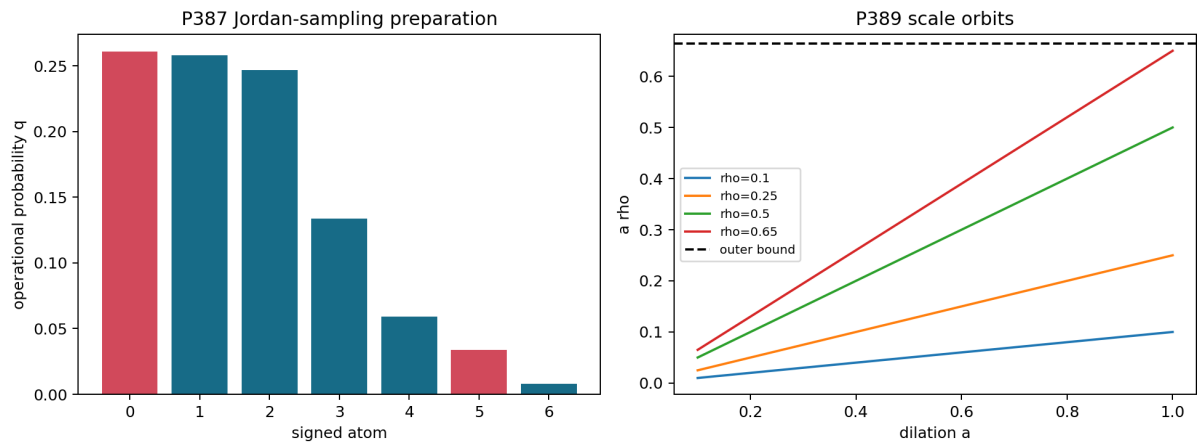


Figure 10.1: Positive Jordan-sampling preparation probabilities and the noncanonical outer-scale dilation orbits.

Chapter 11

P388 — conversion is not implied by co-listing resources

Let

$$\mathbf{r} = (r_1, \dots, r_5) \in \mathbb{N}^5.$$

For every ordered pair $i \neq j$, the generator

$$e_i = (0, \dots, 1, \dots, 0)$$

has $r_i > 0$ and $r_j = 0$. Hence the minimal signature cannot prove

$$r_i > 0 \implies r_j > 0.$$

All twenty directed cross-implications fail.

Identity conversions exist. Reset maps destroy resources. Catalysts cannot create new comparisons because

$$\mathbf{a} + \mathbf{c} \geq \mathbf{b} + \mathbf{c} \implies \mathbf{a} \geq \mathbf{b}.$$

This is not a universal physical no-go. A concrete coupling functor can add conversions, but the coupling must be an additional theorem or axiom.

Chapter 12

P389 — why the analytic extension has no preferred scale

For

$$F_\rho(z) = \sum_{d \geq 0} K(d) \rho^d z^d,$$

define

$$(D_a F)(z) = F(az).$$

Then

$$D_a F_\rho = F_{a\rho}, \quad D_a D_b = D_{ab}.$$

The coordinate

$$t = -\log \rho$$

converts multiplication into addition. However, an invariant positive selector would have to satisfy

$$\rho = a\rho$$

for every admissible a , including $a = 1/2$. This forces $\rho = 0$, which is outside the positive family.

Therefore the semigroup provides evolution or coarse-graining order, but not a preferred origin, unit, or interior scale.

Chapter 13

P390–P393 — external gates

Program	Required external object	Current status
P390	Independent QW hold-out, frozen before evaluation	Not admitted
P391	Traceable $(\ell_*, \tau_*, \hbar_*)$ standards package	Not admitted
P392	Physical reservoir process-tomography record	Not admitted
P393	Blinded electroweak conditional-test bundle	Not admitted

Every bundle must include:

- distinct provider, registrar, and analyst;
- raw-record hash;
- calibration hash;
- freeze timestamp;
- unblinding authorization;
- an immutable failure report.

Chapter 14

Newly constructed theoretical objects

Object	Definition	What it resolves		What open	remains
O121 Kernel-Reflected Terminal Certificate	35 exact Lean-decided integer predicates	Terminal	arithmetic trust	End-to-end	algorithm reflection
O122 Near-Contact Bernstein Extremal Pair	Exact dual plus P366 primal	Oscillatory	gap below 10^{-7}	Exact equality and uniqueness	
O123 Global Strict Radial Embedding Theorem	Injectivity of K_{strict} on $\mathbb{N}_0$	Global	graph-distance naturality	Continuous metrics and other kernels	
O124 Damping-Completion Natural Transformation	C_{damp} entrywise radial multiplier	One bridge square		Source of 9/5, phase, amplitude	
O125 Photonic Component Gauge Quotient Chart	$\mathbb{R}_+^{66} \times \mathbb{T}^{66}$	Coordinate periodicity		Global aliases	nonlocal
O126 Heralded Noisy-Comb Sandwich	Parallel lower and adaptive hybrid upper	Quantified noisy discrimination		Exact optimum	
O127 Clock-Tube Discrimination Envelope	Monotone image of a calibrated time interval	Robust prediction		Clock source	
O128 Jordan Sampling Realization Functor	Positive sampling with signed importance score	One complete mathematical A5 instance		Laboratory realization	
O129 Resource Conversion Obstruction Matrix	Twenty generator countermodels	No forced generation		Model-specific couplings	
O130 Outer-Scale Dilation Torsor	Multiplicative ρ -orbit	Semigroup no-go	canonicity	New boundary/normalization law	

Chapter 15

Falsification ledger

Candidate conclusion	Falsification attempt	Verdict
P380 is only floating optimization	Rational freezing, exact dyadic subdivision, Krawczyk primal, Lean terminal predicates	Survives in declared scope
Strict distance values may collide at large d	Lindemann–Weierstrass contradiction	Refuted globally
Damping completes legacy into strict	Full-profile residual on C_{12}	Refuted
Every photonic phase shift is physical	Independent 2π lattice shifts	Refuted; exact gauge
Coordinate audit proves global identifiability	Distant compensating aliases not classified	Not established
P371 survives noise unchanged	Loss/dephasing sandwich differs from ideal sine law	Refuted
Clock uncertainty is negligible	Explicit tube propagation	Quantified, not erased
Signed mass requires negative probability	JSR positive probability construction	Refuted
Five resources automatically generate each other	Twenty generator countermodels	Refuted in minimal category
Analytic semigroup selects one ρ	Dilation-invariant selector test	Refuted
Code can close external physics	Custody and calibration requirements	Refuted

Chapter 16

Current scientific interpretation

The strongest defensible interpretation after P379–P393 is:

FIN is a dimensionless spectral-operator and signed-resource framework with a globally faithful strict radial graph geometry, nearly closed exact signed-moment certificates, conditional noisy dynamics, and at least one explicit positive-probability operational realization. It still lacks a calibrated empirical realization and internal dimensional standards.

The previous realization gap has split into two layers:

$\mathcal{R}_{\text{math}}$: one FIN signed resource $\longrightarrow$ preparation, free channels, instrument, score, record schema,

which P387 constructs, and

$\mathcal{R}_{\text{lab}} : \mathcal{R}_{\text{math}} \longrightarrow$ calibrated apparatus and independent raw records,

which remains external.

No amount of exact spectral calculus can replace the empirical content of $\mathcal{R}_{\text{lab}}$.

Chapter 17

Recommended Programs P394–P410

17.1 P394 — end-to-end proof-assistant reflection

Formalize:

- rational interval arithmetic;
- fifth-root bracketing;
- Taylor/Lagrange cosine bounds;
- de Casteljau subdivision;
- Krawczyk inclusion;
- weak duality.

Success means the proof assistant derives the terminal predicates rather than only checking their integer forms.

17.2 P395 — exact contact geometry and uniqueness

Use the 3.05×10^{-8} gap to identify the exact active contact pattern. Attempt:

- interval isolation of all dual extrema;
- complementary-slackness equations;
- uniqueness of the signed extremizer;
- an algebraic or transcendental obstruction to exact finite contact.

17.3 P396 — formal global injectivity

Import or formalize the required Lindemann–Weierstrass corollary and prove P381 in a proof assistant. Separate:

- algebraicity of $d^{9/5}$;
- distinctness of the four exponents;
- exponential linear independence;
- categorical corollary.

17.4 P397 — source audit for C_{damp}

Do not repeat the generic bridge. Ask one question:

Can a declared monoid/RG/coarse-graining law generate $(1 + 0.01d)/(1 + d^{9/5})$ without reading the strict target?

Prove a source theorem or a finite no-go for the declared provider class.

17.5 P398 — global photonic quotient identifiability

Classify all parameter pairs in the canonical chart that produce the same complex transfer matrix. Determine:

- the exact gauge group;
- distant aliases;
- singular strata;
- the minimum multitime calibration design.

17.6 P399 — calibrated photonic pilot

External program. Require complex transfer tomography, detector calibration, raw timestamps, a frozen protocol, custody separation, and one-shot evaluation.

17.7 P400 — exact noisy adaptive-comb optimum

Formulate the heralded loss/dephasing process as a finite comb SDP. Produce:

- feasible primal strategy;
- dual upper certificate;
- quantified gap;
- a theorem identifying any regime where GHZ or product probes are optimal.

17.8 P401 — full twelve-mode noisy dynamics

Lift P385 from the extremal two-mode subspace to:

- all twelve modes;
- component loss;
- dephasing;
- detector confusion;
- clock uncertainty.

Distinguish a physical channel from postselected conditional statistics.

17.9 P402 — clock-robust design with a supplied anchor

Optimize sample times against the complete P386 prediction tube. The objective should maximize worst-case discrimination after calibration, not merely minimize interpolation error.

17.10 P403 — executable Jordan-sampling experiment

Turn O128 into an executable specification:

- preparation probabilities;
- atom encoder;
- sign register;
- randomization audit;
- event schema;
- estimator;
- preregistered confidence intervals;
- raw-record validator.

This remains a hypothesis of laboratory realization until physically run.

17.11 P404 — natural transformations among operational semantics

Use the five P373 operational categories, not only their grades. Classify typed functors that preserve preparations, channels, instruments, and monotones.

17.12 P405 — sign-to-phase or sign-to-asymmetry coupling

Test whether the JSR sign record can couple naturally to:

- dephasing coherence;
- reflection asymmetry;
- the P2718 chiral marker.

Reject any construction that merely relabels the sign as a strict selector.

17.13 P406 — canonical scale with one new boundary law

Add exactly one candidate:

- normalized semigroup generator;
- RG boundary condition;
- apparatus resolution;
- entropy reference cell.

Determine whether it selects ρ , and state explicitly whether the result is strict, conditional, or target-defined.

17.14 P407–P410 — external falsification

- P407: independent QW hold-out;
- P408: dimensional standards package;
- P409: physical reservoir process tomography;
- P410: electroweak blind test.

Repository code may validate manifests but may not manufacture evidential status.

17.15 Recommended immediate order

$P394 \longrightarrow P395 \longrightarrow P396 \longrightarrow P400 \longrightarrow P403$.
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Chapter 18

Final conclusion

P379–P393 produces three theorem-level advances.

First, the oscillatory signed-resource problem is now nearly closed with a proof-kernel-checked terminal certificate gap below 10^{-7} .

Second, strict radial geometry is globally faithful on integer graph distances. The earlier quotient category is not merely finite-range evidence: it is the ordinary isometric category for every finite integer-distance carrier.

Third, one previously abstract bridge has been constructed explicitly. Jordan Sampling Realization converts the signed measure into positive preparation probabilities, free Markov dynamics, an instrument, a score, and a record schema. This establishes mathematical operational realizability without pretending that signed mass is negative probability.

The corresponding laboratory bridge remains absent. Dimensional standards, detector calibration, custody, and external records cannot be generated by the spectral theorem or by repository code.

The deepest surviving interpretation is therefore:

FIN has matured into a rigorous dimensionless spectral and operational resource framework with global strict graph geometry and one explicit operational realization. Its remaining transition to physics is an empirical-calibration problem plus the independent selector and dimensional source obstructions, not another identity inside the functional calculus.

Chapter 19

Reproducibility

Run:

- `MPLCONFIGDIR=/tmp/mpl-fin-1033 python3 fin_programs_379_393.py`
- `MPLCONFIGDIR=/tmp/mpl-fin-1033 python3 -m unittest -v test_fin_programs_379_393.py`

Expected result: 16 tests passed, zero failed. Both Lean 4.28 files must compile. P384 and P390–P393 remain external admission gates.

Chapter 20

Suggested citation

Żuchowski, K. (2026). *FIN Research Programs P379–P393: Proof-Reflected Certificates, Global Radial Injectivity, Noisy Dynamics, and a First Explicit Operational Realization* (Release 10.33; Version 1.0.0) [Preprint]. Zenodo.

Reproduction certificate

```
MPLCONFIGDIR=/tmp/mpl-fin-1033 \  
python3 fin_programs_379_393.py
```

```
MPLCONFIGDIR=/tmp/mpl-fin-1033 \  
python3 -m unittest -v test_fin_programs_379_393.py
```

Expected result: 16 tests passed, 0 failed. Both Lean 4.28 sources must compile. P384 and P390–P393 remain external admission gates; no synthetic record is accepted as hardware, standards, hold-out, reservoir, electroweak, or custody evidence.

Suggested citation:

Żuchowski, K. (2026). *FIN Research Programs P379–P393: Proof-Reflected Certificates, Global Radial Injectivity, Noisy Dynamics, and a First Explicit Operational Realization* (Release 10.33; Version 1.0.0) [Preprint]. Zenodo.